Muhammad Mahasin Irfan

Computer Systems and Machine Learning

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Research Summary

I am an undergraduate researcher at LUMS working on memory systems and efficient machine learning. My current work asks when processing-in-memory (PIM) helps an edge system. I help build a sizing tool that combines measurements on real GPUs with PIM simulation to compare hardware configurations and ML workload schedules. I have also built and evaluated quantization tools for diffusion models and studied how evaluation metrics can misrepresent continual model updates.

Research Interests

Computer systems; memory architecture; heterogeneous computing; machine learning systems; performance measurement.

Education

Lahore University of Management Sciences (LUMS) Lahore, Pakistan

B.S. in Computer Science; CGPA: 3.85/4.00 *September 2023 – May 2027 (expected)*

- Dean's Honor List, 2023–24, 2024–25, and 2025–26.
- Selected coursework: Operating Systems; Computer Systems; Networks and Systems; Data Structures; Foundations of Artificial Intelligence and Machine Learning.

Publications

Accepted

- **M. M. Irfan**, M. S. Hassan, and F. Zaffar. "Decomposing Error in Continual World Models." NeurIPS 2026 Workshop on Continual World Models, Idea Track. **Accepted** First author. [Code]

Under review

- Humza Ikram, **Muhammad Mahasin Irfan**, Muhammad Abdullah, Muneeb Pervez, Hashim Sharif, Akash Kothari, Yifan Zhao, and Muhammad Fareed Zaffar. Manuscript on characterizing memory-resident acceleration for edge inference. Submitted to **MLSys 2027**. **Second author.**

Research Experience

Decomposing Error in Continual World Models July 2026 – Present

First-Author Researcher, LUMS; NeurIPS 2026 Workshop on Continual World Models (Idea Track) *Lahore, Pakistan*

Supervised by Dr. Fareed Zaffar (LUMS)

- Derived an exact decomposition of post-update prediction error into inherited error and update mismatch, showing why endpoint accuracy alone may mis-rank continual model updates.
- Demonstrated opposite endpoint/mismatch rankings in 544/544 controlled one-step trials; the reversal persisted in 93.8% of trials at horizon 5 and 66.2% at horizon 12.
- Reproduced the phenomenon across 500 learned-model runs on a nonlinear damped-pendulum simulator; reversal frequency fell from 95% to 6% as inherited error decreased from 0.038 to 0.004.
- Developed the mathematical formulation, implemented the experiments, analyzed the results, and wrote the first-author manuscript. [Code]

PIM Sizing for Edge ML Inference November 2025 – Present

Second-author researcher, LUMS; MLSys 2027 submission *Lahore, Pakistan*

Supervised by Dr. Fareed Zaffar (LUMS), Dr. Hashim Sharif (PhD, UIUC; AMD Research), and Humza Ikram (CMU PhD student)

- Helped develop a tool that compares PIM channel counts, added versus converted channels, chip designs, command-issuing arrangements, and GPU/PIM schedules for latency and energy per inference.
- Built repeatable measurement and experiment pipelines across an AMD Ryzen AI laptop and an NVIDIA Jetson Orin Nano, using deployment-style Triton and TensorRT kernels, scripts, and Docker environments.
- Extended the Newton PIM model in Ramulator using C++ to study bank groups, row sizes, and activation timing; contributed to LPDDR5/LPDDR5X PIM simulation. Earlier configurations improved simulated PIM throughput by 1.3× and 1.5× over the original model. [Code]
- Characterized LLM generation, vision, CNN, and robot-control workloads on two edge systems. The paper projects up to 13.6× faster LLM generation with simulated PIM, while vision workloads gain little; the useful PIM configuration depends on the workload and host system.
- Earlier prototyped a prediction-guided split search that evaluated up to 7–8× fewer candidates and reached within 1% of the measured best latency in selected cases. Nonmonotonic TensorRT curves motivated the submitted paper's measurement-based search instead of relying on that heuristic.

Quantizing Diffusion Models June 2025 – August 2025

AI Optimization Intern, 10x Engineers *Lahore, Pakistan*

Supervised by Dr. Saad Bin Nasir (10xEngineers; formerly Qualcomm) and Dr. Khurram Usman Mazher (formerly Qualcomm)

- Extended QuantX for simulated post-training quantization of Stable Diffusion 1.5 and SDXL, with exploratory SD3.5 support; added component-aware traversal and configurable linear and convolutional quantizers. [Code]
- Built calibration and activation-statistics tools across denoising steps and implemented SmoothQuant-style equivalent scaling.
- Diagnosed catastrophic W4 degradation: excluding the convolution path reduced LPIPS from 0.5713 to 0.3380, while subsequent smoothing improved it to 0.3357 under the recorded evaluation protocol.
- Compared TorchAO, Quanto, and QuantX, and checked how sample count, dataset, compilation, prompts, and random seeds changed apparent image-quality and latency results.

Academic Service

Reviewer, NeurIPS 2026 Workshop on Continual World Models September 2026

Selected by the workshop Technical Program Committee

- Reviewed a workshop submission, assessing what the paper demonstrated and where it needed more evidence, and submitted a written review and recommendation to the committee.

Teaching Experience

Teaching Assistant, Computer Systems and Assembly Language (CS225) Fall 2026

LUMS *approx. 270 students, two sections*

- Develop quizzes, assignments, and a new systems laboratory; grade examinations and conduct tutorials.

Teaching Assistant, Foundations of Data Science Spring 2026

LUMS *approx. 200 students*

- Designed introductory Python and data-visualization assignments, automated tests, and Jupyter/Otter-Grader workflows; held office hours, supported course projects, graded work, and invigilated examinations.

Teaching Assistant, Introduction to Programming (CS100) Spring 2026

LUMS *approx. 180 students*

- Created assignments, held programming labs and a tutorial, and provided student-facing office hours.

Honors and Awards

- **Dean's Honor List**, LUMS: selected in every completed academic year (2023–24, 2024–25, 2025–26).
- **National Mathematics Talent Contest**: top 50 chosen from 4000+ candidates.
- **Purple Comet! Math Meet**: first-place team in Pakistan.
- **Southeast Asian Mathematical Olympiad (SEAMO)**: Gold Medal; third nationally in Pakistan.
- **Full A-Level Scholarship**: 100% tuition scholarship.

Technical Skills

Programming:	C++, Python, C; Bash; Git; Linux; Docker; Jupyter; LaTeX.
ML systems:	PyTorch; TVM; TensorRT; model quantization; diffusion models; performance evaluation and autotuning.
Architecture:	Ramulator/Newton PIM; DRAM architecture; heterogeneous GPU/PIM execution; memory-system simulation.
Research methods:	Empirical evaluation and benchmarking; semi-structured interviews; reflexive thematic analysis.
Languages:	English, Urdu, Punjabi.